SOFTWARE ENGINEERING LAB – EXPERIMENT 5

UML Class Diagram – Project-Based Student Template

Project Title: ***Shield AI***

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Aim

To identify the major classes of our project and prepare a UML class diagram with attributes, methods, relationships and multiplicities.

Shield AI is a text-based scam detection system that accepts text copied from emails, SMS messages, and other sources. It preprocesses the text, checks for scam-related indicators, assigns a risk level, and presents the result with a warning and explanation. The design can also support scan history and user feedback if these features are included in the implementation.

Shield AI analyzes text copied from emails, SMS messages, and other sources to identify potential scam indicators. It preprocesses the submitted text, estimates a risk level, and displays a warning and explanation to help users make safer decisions.

Step 2 – Candidate Classes

Nouns/entities identified from the project's requirements, with a decision on whether each needs its own data or behaviour:

Possible Class	Why is it needed?	Keep? (Yes/No)
User	Represents a person submitting text and reviewing scan results.	Yes
ScanRequest	Stores the submitted text, source type, and scan timestamp.	Yes
TextPreprocessor	Cleans and normalizes input text before analysis.	Yes
ScamAnalyzer	Checks scam indicators and calculates a risk score.	Yes
ScanResult	Stores risk level, explanation, and warning shown to the user.	Yes
ScanHistory	Organizes previous scan results for later review, if history is implemented.	Yes
Feedback	Stores user feedback about a detection result, if feedback is implemented.	Yes
Screen / Widget classes	UI presentation components rather than core domain entities.	No
Email / SMS provider integration	External source or integration rather than a core domain class for pasted-text scanning.	No

Step 3 – Attributes and Methods

The six classes kept from Step 2, with their key attributes and methods:

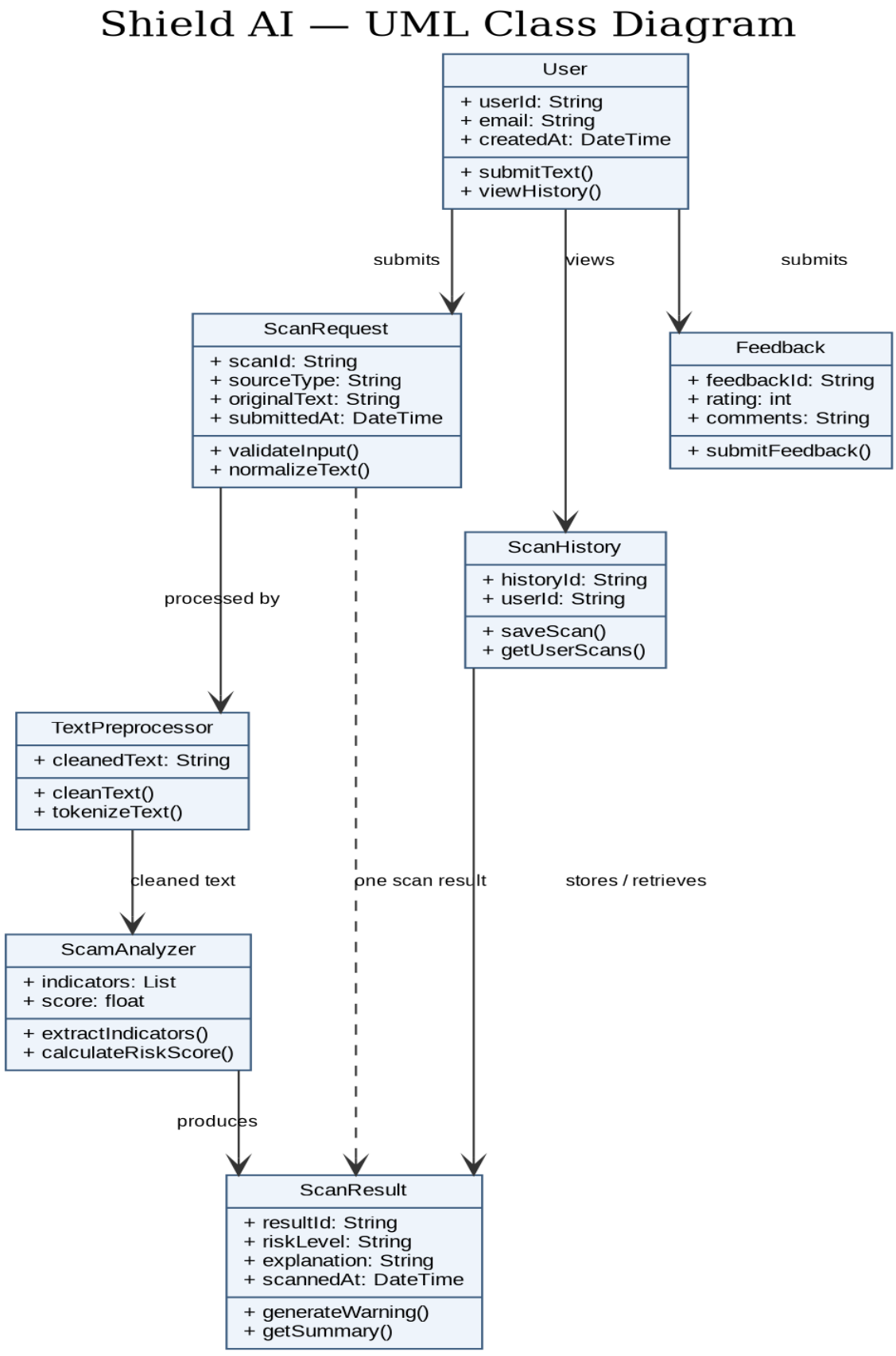
Class Name	Attributes (2–5)	Methods / Operations (1–4)
User	userId: String email: String createdAt: DateTime	submitText(): ScanRequest viewHistory(): List
ScanRequest	scanId: String sourceType: String originalText: String submittedAt: DateTime	validateInput(): bool normalizeText(): String
TextPreprocessor	cleanedText: String tokens: List	cleanText(): String tokenizeText(): List
ScamAnalyzer	indicators: List riskScore: float	extractIndicators(): List calculateRiskScore(): float
ScanResult	resultId: String riskLevel: String explanation: String scannedAt: DateTime	generateWarning(): String getSummary(): String
ScanHistory	historyId: String userId: String scanIds: List	saveScan(): bool getUserScans(): List
Feedback	feedbackId: String resultId: String rating: int comments: String	submitFeedback(): bool

Step 4 – Relationships and Multiplicity

Class A	Relationship	Class B	Multiplicity at A	Multiplicity at B
User	Association — submits	ScanRequest	1	*
ScanRequest	Composition — processed by	TextPreprocessor	1	1
TextPreprocessor	Association — supplies cleaned text to	ScamAnalyzer	1	1
ScamAnalyzer	Association — produces	ScanResult	1	1
User	Association — views	ScanHistory	1	0..1
ScanHistory	Association — stores/retrieves	ScanResult	1	*
User	Association — submits	Feedback	1	*

*Multiplicity reminder: 1 = exactly one, 0..1 = zero or one, * = many, 1..* = one or many.*

Step 5 – Final UML Class Diagram



Quick Self-Check

☑ My diagram matches my project title and requirements.

☑ I selected only meaningful classes.

☑ Important classes have attributes and methods.

☑ Relationships are shown clearly.

☑ Multiplicity is written where needed.

☑ Inheritance is used only for a true IS-A relationship (not used here — all relationships are association/composition, no IS-A case in this domain).

☑ The final diagram is readable and I can explain my choices.

In this experiment, we identified the major classes for Shield AI and prepared a UML class diagram showing their attributes, methods, relationships, and multiplicities.\

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